Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform indentation is maintained throughout the project
2	Proper File Structure	Files and folders are logically organized	Yes	Files are organized into templates, static, dataset, and model files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as `life_expectancy`, `gni_per_capita`, and `predicted_hdi` clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as `predict()`, `load_model()`, and `generate_graphs()` are descriptive.
5	Code Comments	Inline and block comments explain log	Yes	Important sections include comments for better understanding.
6	Modular Design	Code is split into reusable functions/ modules	Yes	The project is divided into `app.py`, `model.py`, templates, and static resources.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code has been removed, and reusable functions are used
8	Error Handling	Exceptions and errors are handled gracefully	Yes	User input validation and exception handling are implemente

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Prediction Module	Python (Flask)	Prediction Page	High
2	Machine Learning Model	Scikit-learn	Prediction Process	High
3	Data Preprocessing Module	Pandas, NumPy	Model Training	High
4	Graph Generation Module	Matplotlib, Seaborn	Analysis Page	High
5	HTML Templates	HTML, CSS, Bootstrap	Home, Prediction, Analysis, About Pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for templates, static files, dataset, and model.
Readability	5	Clear variable names, proper indentation, and descriptive function names improve readability.
Reusability	5	Modular functions and reusable components simplify maintenance and future enhancements.
Documentation / Comments	5	Key sections are documented with comments and project documentation is provided.
Overall Score	4.8	The code is well-structured, maintainable, readable, and suitable for future development